

Meta Flow Experience: Exploring Group Mindfulness Meditation in the Immersive Environment with Multi-senses Feedback

Jih Hsuan Peng
National Taipei University of
Technology
Taipei, Taiwan

Ping-Hsuan Han
National Taipei University of
Technology
Taipei, Taiwan

Cheng-Ta Yang
Taipei Medical University
Taipei, Taiwan

Tung-Hua Yang
China University of Technology
Taipei, Taiwan

Ze-Song Chen
China University of Technology
Taipei, Taiwan

Yi Min Huang
Taipei Medical University Hospital
Taipei, Taiwan

ABSTRACT

This research presents an exploratory study focusing on developing the immersive mindfulness meditation group experience. The primary goal is to observe the impact of Extended Reality (XR) on enhancing people's awareness and cultivating inner tranquility. Previous works have explored the benefit of group meditation and the potential of utilizing multisensory feedback in XR. Our study designs an activity that employs interactive digital art to help the therapist verbally convey the concept of mindfulness meditation through participants' senses, including visual, olfactory, gustatory, auditory, haptic, interoceptive, and proprioceptive perceptions. Unlike the traditional mindfulness meditation experience, the XR experience provides multiple channels for participants to engage in mindfulness, extending beyond mere imagination. By surrounding the therapist's healing script in immersive projection, the experience can stimulate participants in the space in which they exist physically. To collect participant feedback, we did a short questionnaire with 68 responses. The result shows that few attendees would connect the activity with other traditional psychological therapy activities they attended, but most participants indicated that they had a different experience. Like the conventional activity, it brings them a calm and peaceful mind. The study gathered considerations for designing group mindfulness meditation, including participant preparation and selection, the provision of multisensory stimulation, and essential factors when collecting feedback.

CCS CONCEPTS

• Human-centered computing → Mixed / augmented reality.

KEYWORDS

Extended Reality, Group Meditation, Group Immersive experience, Mindfulness

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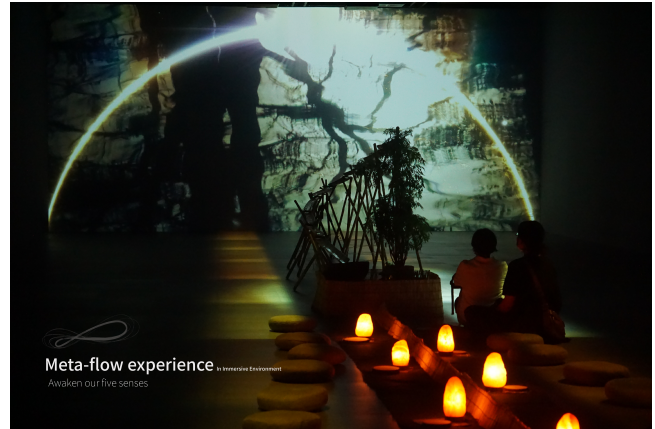


Figure 1: The beginning scene of the Meta Flow experience

1 INTRODUCTION

Functional and structural neuroanatomical data have validated the study of meditation and psychedelic drugs in recent years. It shows that both ways could dramatically affect self-consciousness when the drug or the meditation experience is beyond the threshold, although there are significant phenomenological differences [8]. Besides, the operation mechanisms between those two are also dissimilar. For example, mindfulness meditation is a meditation which experience at the moment without prejudice and judgment and usually needs focused attention and open monitoring practices. In other ways, Psychedelic Drugs mainly act on serotonin receptors in the brain [8]. When it combines meditation and psychedelic drugs, [10] have shown psychedelic drugs to enhance the meditation experience and have positive effects. Conversely, mindfulness meditation helps reduce the anxiety from the drugs. However, the

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danger of hallucinogens is still questionable. It necessitates precise dosage control and is contingent upon individuals' conditions. Therefore, researchers have begun to study whether eXtended Reality (XR) can replace the drugs to create a safer and more effective meditation environment and have had initial results lately. From the result of the Isness [4], the researcher finds there are similarities in phenomenological between virtual reality (VR)-self-transcendent experiences (STE) and psychedelic drugs-STE.

It is worth mentioning that group meditation is sometimes more effective than individual meditation in some cases [7]. Evidence shows that participants who have even a brief group training in meditation could reduce state anxiety after exposure to temporary stress [9]. Based on the above reasons, this exploratory study will focus on observing and discussing whether XR environments can help the experiencer activate multisensory and inner-body perception, enabling them to quickly raise awareness of their bodies and become more focused on themselves.

2 RELATED WORKS

To gain a deeper understanding of mindfulness meditation in human-computer interaction, this section investigates related works about how XR can intervene in meditation practices. As previously suggested by Nina Döllinger et al.[2]. Their study found that most articles reviewed predominantly focus on the recreation and presentation of natural scenes in XR or emphasis the importance of immersion. However, there remains a gap in exploring the connection between participants and the physical environment, encompassing aspects such as their spatial context, interactions with other participants sharing the same space, and the influence of the devices they utilize.

2.1 Individual Meditation with Multimedia

Many study shows that VR is chosen for its increased sense of immersion, like ZenVR[3]. Researchers collected interview data from instructors, expert meditators, and novices. They used the base knowledge from the interview to utilize the interview results to create a VR meditation course system in an undisturbed space. Feedback revealed VR courses helped participants' meditation skill growth, countering concerns about excessive reliance on technology. However, device interference requires learning and adaptation. The Portable Interactive Meditation Mirror[5] offers a unique mindfulness meditation experience by eliminating complete immersion and minimizing the reliance on wearable devices. Its interactive and fun-guided approach targets regular children to learn meditation and reduce stress. However, the study shortage sufficient user feedback and data to compare its impact on meditation in this environment. In XR meditation experiences, visual feedback is prominent. Recent studies have started including other sensory stimuli, TranScen in Stillness [6], which combines olfactory stimulation with VR. In this exploration, participants reported feeling relaxed, but some had difficulty sensing the scent when facing away from the incense burner or distinguishing between different smells. The study couldn't determine if this was due to participants' olfactory senses or a system issue.

2.2 Group Meditation in XR

While VR enhances immersion, it may lead to feelings of isolation. However, Isness [4] explores how to bridge this gap and foster communication among members in a group meditation. The study explores the combination of psychotherapy, psychedelic drugs, and VR to induce 'mystical-type experiences' (MTEs) characterized by connectedness, transcendence, and ineffability. They conducted meditation training in a group using HTC Vive Pro and conductive fabric gloves. Promising results were obtained, with participants reporting MTEs comparable to high-dose psilocybin and Lysergic acid diethylamid studies. The paper inspires VR experiences that foster connections among participants, challenging the perception of isolation in VR. The use of interactive conductive fabric gloves enables participant interaction. While feedback on individual and group experiences is not emphasized, VR phenomenology in a supportive environment may generate insightful MTEs. In [1]'s research, a novel form of group mindfulness meditation was revealed, where a participant's EGG data generated a soundscape for the rest of the members to walk and practice mindfulness meditation in sync with the rhythm. The study showed that the rhythmic soundscape aided participants in experiencing the present moment and inducing relaxation. Based on the above literature, most studies focus on exploring individual meditation experiences and enhancing visual immersion. However, using HMDs and wearable devices requires time to adapt, and the relationship between the meditators and the physical environment has yet to be thoroughly explored. Therefore, our exploratory research is centered on group mindfulness meditation experiences without needing wearable equipment. Additionally, we observe participants' responses to stimuli beyond visual cues and collect their feedback.

3 META FLOW EXPERIENCE

The Meta Flow Experience is designed to create an immersive environment for exploring group mindfulness meditation with Multi-senses Feedback, using five-wall projections and physical devices.

3.1 Arrangement of Physical Environment

Shown in Fig. 2. The space where the experience was presented is an (L)18.46m x (W)8.65m x (H)2.5m rectangular room. Our system employed projectors to achieve a 360° surround with floor projection, enhancing the feeling of immersion, along with sixteen stereophonic speakers for surround sound, creating a dynamic environment encompassing visual and auditory sensations. In the center of the space, a water circulatory system comprising ten two-meter bamboo structures forms a bamboo accents water fountain, a traditional Japanese water feature. It connects to a water tank that covers a specially designed plant island, concealing the water pump. This setup creates a cozy environment and provides gentle stimulations of visual, olfactory, auditory, and haptic perception. The remote-controlled salt lights system enabled the researcher to regulate the on-off frequency, assisting participants in breath training and providing a sense of warmth and negative ions. Furthermore, the space was equipped with soft cushions for participants to rest, and wooden mats with paper cups containing goji berries were placed in front of them. As shown in Fig. 3. Our work consciously

integrated diverse materials to offer participants a range of sensory experiences.

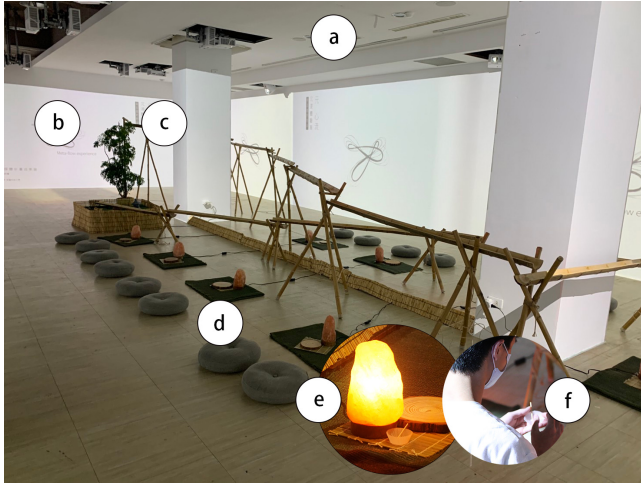


Figure 2: Meta Flow Experience is a group mindfulness meditation immersive experience with MR and multiple senses. Following are the scenes we designed: (a) a recorded guiding voice from the therapist, (b) a 360° video, (c) a bamboo accents water fountain, (d) soft cushions (e) a remote-controlling salt lights system, (f) goji berries.

3.2 Multiple Sensory Stimuli

The system provides multiple sensory stimuli, as shown in Fig. 4. We use Unity to provide 360-degree panoramic projection, presenting a seamless fusion of macro photography, urban night scenes, outer space, forests, underwater scenes, deserts, and various other scenes in the digital world. Combined with the physical space's salt lights and venue decorations, it offers participants immersive visual experiences spanning different worlds. Participants can experience gentle background music and prerecorded therapist-guided instructions through the surround sound system. They can also hear the actual sounds of flowing water and the striking of singing bowls. In addition, they may unavoidably hear ambient sounds from the physical environment, such as footsteps, breathing sounds, and even whispers from other participants. The system incorporates the scent of woody essential oil into the water. Additionally, natural ambient odors, such as those from other participants, plants, bamboo, water, wood, and Goji berries, are not deliberately removed from the environment. Our experience allows participants to taste Goji berries, a sweet-tasting Chinese herbal ingredient. Generally, Goji berries have a mild flavor and a texture similar to raisins. Participants can decide whether to taste them. In this experience, various materials provide participants with diverse haptic sensations, most of which come from the natural environment, such as fabrics, water, wood, bamboo, stones, and leaves. The installation of salt lamps releases negative ions into the air, which may not be easily detectable by the participants. Following the voice guidance, participants are guided to keep moving during certain segments, giving them more stimuli originating from within themselves.



Figure 3: The stimuli provided by the Meta Flow Experience include (a) visual stimuli consisting of digital imagery and environmental setups. (b) olfactory stimuli, with woody essential oils. (c) gustatory stimuli, offering each participant a Goji berry. (d) Goji berries. (e) tactile stimuli, providing various textures, including heat perception.

3.3 Experience Script

The script's objective is to lead the participants through the complete process of experiencing mindfulness meditation. The experience commences with paying attention and accepting yourself, as a brief introductory voice reminds the participants to relax and focus on living in the moment. Utilizing micro-visuals and background music, the participants gradually become oriented and enter an almost dark space to commence breathing exercises and body scan meditation. Subsequently, the guiding voice prompts them to open their eyes and observe the projected urban night scenes the participants are familiar with while tasting the Goji berries before them. The scene then transitions to outer space, with the guiding voice encouraging participants to stand up and explore the present space. In the following step, the guiding voice instructs participants to engage in mindful walking along the guided light from the XR source and salt light, traversing underwater, desert, and forest spaces and experiencing the warmth of walking and other sensory stimulations. Finally, the participants transition into sitting meditation, returning to a seated position in the forest, where they redirect their focus to themselves, feel the bodily sensations from the experiences, and conclude the entire journey. Throughout the execution of the script, the projection will be synchronized to showcase various scenes. The bamboo-accented water fountain and remote-controlled salt lights system will also be activated or deactivated accordingly.

4 ACTIVITY PROCESS AND PARTICIPANT

The first thirty minutes were open for participants to sign-up and look around the surroundings. The projection only showed the activity's title instead of the 3D video. They can see and touch the decorations or talk to others. There was staff to advise if they

needed it. The follow-up was the twenty-five minutes group immersive mindfulness meditation experience in the closed space. The researcher could control all the lighting to keep the quality of the projection, and the staff would still be there to assist the attendees. In the end, They would be asked to fill out a survey. Overall, the experience had sixty-eight attendees; thirty-seven were female. 64.7 percent are around nineteen to twenty-three years old. The median is twenty-one years old. Fifty-seven had yet to experience this kind of activity, and Forty-two people joined the experience without accompanying.

5 RESULTS AND DISCUSSION

Our system provides an environment for observing participants as they experience mindfulness meditation in an immersive virtual environment without wearing any interactive devices. The study investigated how audiences sensed during the experience. Most participants did not experience a similar activity. However, eight participants denoted they had. Two connect the activity with an immersive exhibition created by TeamLab, an international Japanese art collective, and others link the experience with the traditional meditation they attended. Two attendees claimed they had different experiences that clearly distinguished traditional meditation, an immersive digital art exhibition, and this XR activity.

When we asked about the most impressive moment during the activity, thirteen people answered eating goji berries, and nine answered the sound of running water. Six people think the cosmic scene was impressive. Five attendees think the keeping walk around and when they did the "meditation" was the most memorable part of the experience. The last question we asked about is the overall feeling that our activity brings to you. Nearly all attendees provided positive feedback on the experience. Thirty-eight participants felt a sense of overall calmness, physical and mental relaxation, and a general feeling of well-being after the activity. Three individuals mentioned that the system efficiently facilitated their entry into a meditative state. Additionally, twelve attendees reported experiencing a sense of self-awareness or a connection with their inner selves. On the flip side, five participants noted that the image quality hindered the immersive experience.

5.1 Compared to Traditional Meditation

Although the experience has received positive feedback, such as calmness, relaxation, and well-being, and participants have derived benefits from it, the experimental results could not confirm that the Meta Flow Experience conveys the same phenomena experienced by a traditional one. Unlike previous studies exploring the effects of hallucinogens on mindfulness meditation, this study did not demonstrate the positively experienced self-dissolution based on the feedback received. Only six people associated it with their experience in the past. The lack of comparison may be attributed to the fact that most participants had yet to gain prior experience with mindfulness meditation.

5.2 Feedback on Various Stimuli

In this activity, participants were exposed to various stimuli. For some individuals, these stimuli contributed to an overall mindfulness meditation experience, such as walking meditation and visual

stimuli from the panoramic images. Additionally, certain stimuli appeared more prominent than others, such as eating goji berries and listening to flowing water, which left a deep impression on many users. Still, whether these stimuli enhanced or weakened the meditation connection was uncertain. Through observation, it was noted that some participants exhibited notably similar responses to certain senses. However, the study could not be determined whether this was attributed to the design of the experiential system or certain senses being more easily awakened.

5.3 Group Meditation with Multimedia

Throughout the experiential activity, participants were free to move around and interact with others for a considerable amount of time. However, based on our open-ended questionnaires, the presence of other individuals in the same room did not impact the individual's experience. This phenomenon may be attributed to the guidance provided by the facilitator, which did not encourage people to interact with each other.

6 LIMITATION AND CHALLENGES

This activity utilized projectors and speakers to create a changeable immersive space, avoiding participants needing to adapt the wearable devices and allowing them to move around freely. However, participants seemed to be more susceptible to disruptions in the physical environment when not using any wearable interactive devices. These disturbances may arise from external sources, such as external sounds or lighting, and internal setup elements, like exposed equipment and visible wires. Additionally, as the activity cannot ensure every participant is in the same location, providing equal stimuli to all individuals in the group becomes challenging. For instance, some may be closer to the speakers, while others may be nearer to the water source or scents.

7 SUMMARY AND FUTURE WORK

In previous works, the multimedia system for meditation research has primarily focused on VR and visual stimuli. In contrast, our designed system aims to create a virtual, authentic, intertwined, immersive, and multisensory group meditation experience for participants without wearable devices. The goal is to observe and explore whether participants can achieve a positive group meditation experience in such an environment. Based on the participants' positive responses. Our work will consider these observational findings in future studies and refine the configuration to explore various sensory thresholds. We plan to incorporate devices that capture participants' bodily signals and eliminate reliance on subjective questionnaires alone. Moreover, we will categorize participants into experienced and inexperienced groups, allowing for comparisons between traditional meditation and our approach or measuring the extent of learning mindfulness meditation.

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